



# Mohammed Siddiq

AI ML Engineer • Generative AI & NLP • Python & SQL • MLOps

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Open to Relocate: Dubai, UAE | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

## PROFESSIONAL SUMMARY

AI/ML Engineer with a B.Tech in Artificial Intelligence & Machine Learning, AWS Cloud Practitioner certification, and 1+ year of enterprise technical operations experience. Skilled in Python, SQL, NLP, Generative AI, RAG, LangGraph, and MLOps, with hands-on experience deploying containerized AI applications on AWS. Open to relocating to Dubai, UAE for AI/ML and Python engineering roles.

## EXPERIENCE

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**Technical Operations Engineer | Tech Mahindra**

**Aug 2025 – Sep 2026 • Hyderabad, India**

### US-Based Enterprise Client Engagement

- Resolved 10–12 enterprise incidents daily across 2,000+ retail locations using multi-table SQL queries, root cause analysis, and structured operational data.
- Investigated recurring system failures by analyzing data from multiple enterprise platforms, enabling proactive maintenance and improved SLA adherence.
- Collaborated with a 30+ member operations team and field technicians to coordinate escalated technical issues and minimize site downtime.
- Maintained technical documentation, configuration records, and incident history to ensure auditability and operational continuity.

**Machine Learning Intern Ignitus**

**Feb 2024 – Jul 2024 • Remote**

- Developed predictive modeling and NLP solutions using Python, Pandas, and Scikit-learn, applying feature engineering to improve model performance.
- Engineered end-to-end data pipelines and ETL workflows to clean, transform, and preprocess structured and unstructured data for machine learning applications.
- Optimized ML inference and data processing pipelines through code refactoring, improving execution efficiency and application performance.
- Collaborated in an Agile development environment using Git, feature branches, pull requests, and peer reviews to build and integrate 3+ reusable ML components for testing and deployment.

**Data Analyst & Web Developer (Contract) | Excel Placement Services**  
**India**

**Oct 2024 – Mar 2025 • Hyderabad,**

- Analyzed recruitment data using SQL, Excel, and Power BI, building KPI dashboards for Time-to-Hire, Submission-to-Interview, and recruitment pipeline performance.
- Automated reporting workflows using Excel, streamlining recruiter reporting processes and improving operational productivity.
- Designed, developed, and deployed a responsive full-stack recruitment website, managing the complete project lifecycle from requirements gathering and UI development to production deployment, contributing to a 40% increase in client acquisition inquiries.
- Led the end-to-end website migration by configuring GoDaddy hosting, DNS records, domain setup, backend components, and production deployment, while optimizing application latency from 1–1.5 seconds to under 500 ms with minimal downtime.

## PROJECTS

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### **Masar — Dubai Congestion Prediction & Commute Recommendation System**

[Masar](#)

- Engineered and deployed a live-data AI mobility platform using **Python, LangGraph, XGBoost, NetworkX, Docker, Nginx, AWS EC2, DuckDNS, and Let's Encrypt**, exposing the application through a production HTTPS endpoint.
- Built a stateful **LangGraph** workflow combining congestion prediction, confidence-based routing, LLM-assisted context reasoning, graph-based route optimization, and bilingual English/Arabic recommendation synthesis.
- Integrated TomTom Traffic API, OpenWeatherMap API, Supabase (PostgreSQL), and Groq LLM APIs to ingest runtime traffic/weather signals, transform live inputs into model features, and generate context-aware commute recommendations.
- Containerized the application with **Docker** and deployed it on **AWS EC2** (t3.micro, Ubuntu 26.04) behind an **Nginx reverse proxy**, configuring DuckDNS DNS, HTTP→HTTPS enforcement, and automated Let's Encrypt certificate renewal with Certbot.
- Implemented production API endpoints for health checks, live traffic retrieval, location queries, and recommendation generation, with environment-based configuration and redacted external API reporting for operational debugging.

**Tech Stack:** Python • LangGraph • XGBoost • NetworkX • Groq API • TomTom Traffic API • OpenWeatherMap API • Supabase • Docker • AWS EC2 • Nginx • DuckDNS • Let's Encrypt

### **RAG-Powered Multi-Document QA System**

[RAG](#)

- Engineered an end-to-end Retrieval-Augmented Generation pipeline with multi-format document ingestion (PDF, DOCX, TXT), semantic vector storage via **ChromaDB**, and advanced retrieval strategies (MMR + MultiQueryRetriever), delivering cited, cross-document synthesised answers at scale using **LLaMA 3.3 70B** via **Groq** API with full source attribution and strategy-switching CLI.

**Tech Stack:** Python • LangChain • ChromaDB • HuggingFace Sentence Transformers • Groq API • LLaMA 3.3 70B

## TECHNICAL SKILLS

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**Languages:** Python, SQL

**ML / AI:** NLP, Deep Learning, Generative AI, LLMs, RAG, LangGraph, XGBoost, Vector Embeddings, Prompt Engineering, Transfer Learning

**Frameworks & Tools:** PyTorch, Keras, Hugging Face, LangChain, Pandas, NumPy, Scikit-learn, Flask, NetworkX

**MLOps & Cloud:** Docker, Docker Compose, GitHub Actions, AWS (EC2, S3, Lambda), REST APIs, Git

## EDUCATION

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### **B.Tech – Artificial Intelligence & Machine Learning**

2020 – 2024

*Siddhartha Institute of Technology and Science, JNTUH*

- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Database Management Systems, Computer Networks

## CERTIFICATIONS

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- AWS Cloud Practitioner Essentials – AWS Training & Certification (Jul 2026)
- Machine Learning Training – Internshala
- Deep Learning with TensorFlow – IBM
- Machine Learning with Python – freeCodeCamp
- Responsive Web Design – freeCodeCamp